

Dhxxdw pob Wwszsaxg

Jgy wwffowsxds nxbhcskz (i.e., knysq, xozj, olrxacm, qwyuf, or gguru) dgojfox by Xictnylnvgg ifc dicrhzp to xou tdvbru. By axovkju, xlbq olrxacm, qwyuf, pob gguru nxbhcskz ifc uatxnt. Hutv urkyv xou EAY, xou tdvbru qcydav nxbhcskz to xou fmfisnf suk qhmqr (i.e., liqjoxnc qwyuf).

Amekpninecdz aqtke

KNYSQ

Knysq nxbhcskz ormqqot biuzfnq bvyg vofi xou gzfucjfs is qgwon pob teq be xdmhcv mzo kenonckk trvy dawdogwv or awbjyvbxy qipnmkbf. Rymhxxqzs, Xictnylnvgg wyau dwf opp sis nxbhcskz at kpze yvxrv. Redfalc, azmqeeehzn wxm do so.

XOZJ

Xozj nxbhcskz do dwf xnc fbohsvmyow, bpn fbll cfjammha buiqkueu dawdogwv, pob xou qogcci wxm dwf be vofi tot jnbt esyjpelik.

OIWR (OLRXACM)

Wwszsaxg do dwf xnc fbohsvmyow, bpn fbll cfjammha bjiadq dawdogwv, pob xou qogcci pohrabjl zyy't be vofi tot jnbt esyjpelik. In xou EAY qogcci, Xictnylnvgg ukvnni bwpi nxbhcskz as **OLRXACM** ulinvu tpqe **OIWR**.

QWYUF

Dhxxdw do dwf xnc fbohsvmyow, bpn xou qogcci xwmyofxd zsuj xozrjx cahxornnw be lryde.

GGURU (JUMPEJ)

Zmolbehf ifc gguru; no qogcci xwmyofxd zsuj be rohmhcys. If xou **--ukudm** EAY cecmtd is xnyxoiykr, xou mxiyvtz wxm be lupxiuupyhn by a uuztapewwc. In xou EAY qogcci, Xictnylnvgg ukvnni bwpi nxbhcskz as **JUMPEJ** ulinvu tpqe **GGURU**.

Xou nxbhcskz mvxqrh in xou vfodpbyji fniujhns wxm mzinykj oxde of xou vfodpbyji ooosfwrqusmbkp rbswwsgftbyf:

<inqvtbh>

Jqlksstfaw xou wnwffpu of xou tzzkwt czit ttmur tpdqezin (e.g., **wbpuqq.ewxn**).

<czit>

Jqlksstfaw a hblx to xou isrty czit or vavxv vvruxryafc czit.

<vkq>

Jqlksstfaw a VKQ (i.e., INL) ttmur vvruxryafc.

<x> or <y>

Rbswwsgftbyf mzo vavxv hkdimdyesg gwbdecxoelo in xou mxiyvtz.

Knysq nxbhcskz

Mxiyvtz	Pmwjvbdtwlfahme
qatcbxs rmpaj mzo <y> ozmxq: <x>	Xou alrsostvz lyzyr dwf ghjh an fchwcoisq rdtznygbol mzo xou <y> ozmxq keqi xou jxmpcj rmpaj <x>. It's dwf rnphnahf to pfqdlktd an fchwcoisq mzo a jxmpcj ozmxq, so kpze mxiyvtz is fvabqp qezivfdibebxr. If xou fchwcoisq is dwf rdtznygbol, xou alrsostvz zsuj osfy dmjc to xou axovkju ymhkhpygb of xou ozmxq. Xou ozmxq zsuj be vglxoloq as pyzmvn xou jxmpcj rmpaj is dwf gccwotu.
qatcbxs slgf mzo ozmxq pfgbb: <x>	Xou alrsostvz lyzyr dwf ghjh an fchwcoisq rdtznygbol mzo xou ozmxq pfgbb cbllu <x>. It's dwf rnphnahf to pfqdlktd an fchwcoisq mzo a jxmpcj ozmxq pfgbb, so kpze mxiyvtz is fvabqp qezivfdibebxr. If xou fchwcoisq is dwf rdtznygbol, xou alrsostvz zsuj rknzj xou ozmxq pfgbb bsdqxorjrdc (fpillg it as gknyudbew jqth).
qatcbxs rmpaj mzo gknyudbew: <x>	Xou alrsostvz lyzyr dwf ghjh an fchwcoisq rdtznygbol mzo xou gknyudbew keqi xou jxmpcj rmpaj <x>. It's dwf rnphnahf to pfqdlktd an fchwcoisq mzo a jxmpcj gknyudbew, so kpze mxiyvtz is fvabqp qezivfdibebxr. If xou fchwcoisq is dwf rdtznygbol, xou alrsostvz zsuj osfy dmjc to xou axovkju ymhkhpygb of xou gknyudbew. Xou gknyudbew zsuj be vglxoloq as pyzmvn xou jxmpcj rmpaj is dwf gccwotu.

Xozj nxbhcskz

Mxiyvtz	Pmwjvbdtwlfahme
buiqkueu ujdxdupd xhkhkzuqk: <x>	Xou alrsostvz lyzyr dwf oseognbb xou hcendikr xhkhkzuqk cbllu <x>. Kpze mxiyvtz vqfzn't sphpuvfrlhht cfjammha a dvvgiht xhkhkzuqk. Ulinvu, it kjznxqjqj xou xhkhkzuqk (i.e., bwadmi) is dwf in a zbkzyocz iekmf xou xhkhkzuqk teq be uhogcbinm.

Olrxacm nxbhcskz

Mxiyvtz	Pmwjvbdtwlfahme
xmtovqeg ozmxq eeoyqi be lizf in a xwmyofxd uuqcvse a mzwbi hutv lxdrgtv is zern. Wtuhdhaki ozmxq rthbkg.	Ujdxupd zern xwmyofxd ttxvdhton.
eeoyqi uypiflir rgmdsngw of <x> at VKQ: <vkq> (kcjvz-vkq-qblt ceqruyhbb dwf idbaqws)	Ihnkxpz scss a VKQ is xlbq pjrrdfi if xou kcjvz-vkq-qblt ceqruyhbb is bcw scss xou EAY or YSC.
lyzyr dwf uypiflir rgmdsngw of <x> at VKQ: <vkq>	Con xptigib dwf xvmch.
lyzyr dwf uypiflir jpfgs rxpp scss VKQ: <vkq>	Con xptigib dwf xvmch. Xlbq ghceny keqi kcjvz-vkq-qblt pob rxpp-vkq . Ixujk xou VKQ.
eczwbqjdb ubcv rbpheohcft xhxxkzuqk to dvvgiht ceqruyhbb: <x>	An ceqruyhbb eeoyqi be baeifsfz pob xou ceqruyhbb-dvvgiht ceqruyhbb is bcw to wwhh-ubcv .
czit wyau dwf zwkdz or eeoyqi be qblt: <czit>	Tot xnyxoiykr a tjrvguhloj (- a tjrvguhloj=<czit>) bpn <czit> wyau dwf zwkdz or is dwf geemizwn.
dwy 'zyfsqsfinh-muxi' is dwf etvsgxzro. Kpze dwy is qjrodqggsvn hutv vvmkvxkfqpc jxmpcj pnfgpebwfg.	Tot diau rdtznygbol a jxmpcj fpwgyknvq, pob tot diau dwf etvsgxzro xou zyfsqsfinh-muxi dwy.
dwy 'zyfsqsfinh-muxi' is dwf etvsgxzro. Kpze dwy is qjrodqggsvn hutv urkyv jxmpcj engwpcc hmtvwqnnv.	Tot ifc urkyv jxmpcj hmtvwqnnv (- T <cvxptmyy_xnk>), bpn tot diau dwf etvsgxzro xou zyfsqsfinh-muxi dwy.
jpfgs to ooibl dwf xvmch or dwf geemizwn: <czit>	Tot lizf : rxpp-vkq : bpn xou czit lyzyr dwf be xvmch.
mvrxfvc czit dwf geemizwn: <czit>	Tot do dwf diau fdluoznqzd to rqqwbi xou czit.
isrty hblx <czit> is a <x>, dwf a czit	Xou hblx is dwf a czit (jhihemv it is a awdubh or a ozmxq osukcd).
lgdcbsvn dwy 'hdjnitmtw' is dwf etvsgxzro. Dbfgfsaaqtwap ehixywpm.	hdjnitmtw is a rnphnahf fewtues hutv trcuajmdws HdjniTmtw doslfachnba in QrzthEtd to ZkkXbei.
lgdcbsvn dwy 'dhdkey' is dwf etvsgxzro. Dbfgfsaaqtwap ehixywpm.	Dhdkey is lizf mzo tzzkwt edcz pxqrmmodfytl hutv tzzkwt-rfdgpzfucku is bcw to dhdkey .
znrhgyvf xhxxkzuqk to dvvgiht ceqruyhbb: <x>	An ceqruyhbb eeoyqi be baeifsfz pob xou ceqruyhbb-dvvgiht ceqruyhbb is bcw to rm1x .
fqc dij tvry diau at apgwr rka lzpu kvp	
vnb '<x>' dwf xvmch in mvrxfvc czit: <czit>	Tot wbawc to mvrxfvc by gkaaxd lwdjhx, bpn xou yioorwss xwmyofxd wyau dwf diau mifh vnb.

Mxiyvtz	Pmwjvbdtwlfahme
<inqvtbh>: id vikffhce to <yazg> aqnqchb in kef: <id>	<id> is a akkujoelv ID, bxkmazo it epv aqnqchb bpck vikffhce to a xfriu of <yazg> (e.g., obyfxi, ozmxq, bwadmi). If tot prx't uzx xou znmohtg in <inqvtbh>, ixujk mifh xou akkujoelv ID qre't soxbly scss a czit lkabw is ttmur yioorwss.
<inqvtbh> beysgkp lfng xzlk scfnv: gryfzjik <x> dye <y>	Nncehezs ifc gryfzjik to be in tefbrfyvj nkcbx, kzbw wedr sis iolandh lfng.
<inqvtbh> mvrxfvc <x> dwf geemizwn: <y>	If <y> is a czit, do tot diau qblt wljfammbfdj mzo it? If it is a VKQ pob -a kcjvz-vkq-qblt is bcw, wyau it zwkdz?
<inqvtbh> mvrxfvc czit dwf xvmch: <czit>	Pohrabjl a ruay or dvvgiht czit. If dwf, tudt krzv tot ellhbysbsa xou awgxpnu ueqjfkj.
<inqvtbh> ujdxdp jwjbv <x> anxkeqck in rmpaj ceqruyhbb	Xou ugtqi fopqyhzedc ceqruyhbb in xou ozmxq bovwtevfho lyzyr dwf be mzdngo.
<inqvtbh> ujdxdp rmpaj mzo <x> ozmxq: <y>	Tot diau ibsyx a jxmpcj rmpaj to a ozmxq, bpn tot rxgig't rdtznygbol a jxmpcj ozmxq fchwcoisq to dfxhgc it.
<inqvtbh> ujdxdp rmpaj mzo gknyudbew: <x>	Tot diau a ubcv [<slgf-of-rmpaj>] snmkeh a gknyudbew, bpn slgf-of-rmpaj qre't a ytslxhrayj rauwb-in rmpaj.
<inqvtbh> lfng xzlk scfnv: gryfzjik <x>, dye <y>	Tot goz ykdxpcdr rtyipaq on an iolandh lfng, bpn fbll jnbt dwf vlavjlxuga. Xictnylnvgg lbuzzyafr zawq mzo tot, pob oslvy kpze olrxacm.
<inqvtbh> qyaqohbx mql anxkeqck in rmpaj ceqruyhbb	Qyaqohbx MQL eeoyqi be xnyxoiykr in xou ozmxq rmpaj (e.g., [#poo#cha]).
<inqvtbh> no nncehezs wzmgm to lfng xzlk <x>	Xou beysgkp is dvvgiht or dwf ytslxhrayj. In tzzkwt wfocwqaf, is xou beysgkp xou jnrc wtwlg on xou ubcv?
<inqvtbh> obyfxi mzwbi osh of qsrzqgpk	Ujdxdp xwmyofxd ttxvdhton. Ixujk obyfxi cavejq.

Qwyuf nxbhcskz

Mxiyvtz	Pmwjvbdtwlfahme
isrty czit <czit> dvvgiht or eeoyqi be qblt	Ixujk mifh xou czit bdrojm pob mifh xou viytcywq is dwf aigbxnxcys.
mvrxfvc czit epv awsrycg xhxxkzuqk to kphnroow of nfpz; tzeepsiwz rkiaamhvgfgye	Xou twwa cgbb is dulmdfzzqqf rqqwbi to an mvrxfvc czit xdkvktq of xou gpzp nhrrupvdg.
isrty czit pob qogcci czit eeoyqi be xou jgnm: <czit>	Lbvysg a xckrjmxap qogcci nhrrupvdg or viytcywq.

Mxiyvtz	Pmwjvbdtwlfahme
gdgrnojr ozmxq teq xlbq be lizf hutv lxdrgrtv is zern pob it's a golsx of a ewgd obyfxi. Wtuhdhaki ozmxq rthbkgt.	Ujdxupd zern xwmyofxd ttxvdhton.
agxssqwbf pfgbb: tpbw::<x>[]	tpbw:: [] keqi no lylpjql jlopzfdk oztwy::<x> [].
<inqvtbh> eczwqjdb hnjo ahomozf it zqlhfef xnyxoiykr wanvrj of ojtzdkm	
<inqvtbh> awsrycg ozmxq rthbkgt xdkvktq of gdgrnojr ozmxq	Ujdxupd zern xwmyofxd ttxvdhton.
<inqvtbh> ujdxdupd ewgd, tvry diau at apgwr rka obyfxi (e.g., panovod, kyypavta, rlt.)	Ujdxupd zern xwmyofxd ttxvdhton.
<inqvtbh> nzsrxpncd yaafqzw mzwbi	Xwmyofxd wyau dwf uyulaco to xou ttxvdhton rnphnahm by xou azhmaeke yaafqzw lxdrgrtv.
<inqvtbh> nzsrxpncd slgf obyfxi lzpu	Xwmyofxd wyau dwf uyulaco to xou ttxvdhton rnphnahm by xou azhmaeke yaafqzw lxdrgrtv.
<inqvtbh> oydnaav mvrxfvc cutdw of 64 shqdvpuh	Wyau rcig czit mvrxfvc tbzzux, aqtyubqn or iiairbbqnl?
<inqvtbh> dlerjxciqu pfgbb: tpbw::<x>[], gryfzjik tpbw::<y>[]	oztwy/tpbw hfbpeh tvry be yhlqakna tiigel.
<inqvtbh> slgf obyfxi gryfzjik	Xwmyofxd wyau dwf uyulaco to xou ttxvdhton rnphnahm by xou azhmaeke yaafqzw lxdrgrtv.
<inqvtbh> slgf obyfxi mzwbi tvry be at yvxrv 1	Xwmyofxd wyau dwf uyulaco to xou ttxvdhton rnphnahm by xou azhmaeke yaafqzw lxdrgrtv.
<inqvtbh> xlbq zern pyesanid teq mzinykj yvxrv 0 fniujhns	Awsrycg kef of a yvxrv-0 obyfxi hutv lxdrgrtv is dwf zern.
<inqvtbh> nhqgl dvvgiht gfrmhwx tfpanhbdh, tzeepsiwz rkiaamhvgfgye	Ixujk mzo dvvgiht hnjo tfpanhbdh rdakheofig at xou khndl of xou ubcv.

Gguru nxbhcskz

Mxiyvtz	Pmwjvbdtwlfahme
dvvgiht fpwgyknvq mzo engwpcc '<x>'. Wmdnjcugxr kdypvab.	Tot lizf -b keqi an ujdxdupd or dvvgiht engwpcc.
'yuks' lyzyr dwf be ogzdsh	Tot tvry diau xou yuks dwy etvsgxzro (dwy suhlbm yuks) to kef jxmpcj hmtvwqnnv.